## 3.3 Main parts of shaping machine

### Base




The base is hollow and is made of cast iron. It provides the necessary support for all the other parts of the machine. It is rigidly bolted to the floor of the workshop.

### Column

It is a box like casting mounted vertically on top of the base. Two accurate guideways are machined on the top of the column. The ram reciprocates on these guideways. The front face of the column is provided with two vertical guideways. They act as guideways for the crossrail. Crossrail moves vertically along these guideways. The column encloses the ram reciprocating mechanism and the mechanism for strokelength adjustment.

### Cross rail

It is mounted on the front vertical guideways of the coloumn. The table may be raised or lowered by adjusting the cross rail vertically. A horizontal cross feedscrew is fitted within the crossrail.

The construction of a shaping machine is shown

### Saddle

The saddle is mounted on the cross rail. It holds the table rigidity on its top. The saddle can be moved in crosswise direction, by holding the crossfeed screw by hand or power.

### Table

It is an important part useful in holding the work firmly on it. It is mounted on the saddle which is located above the crossrail. The top and sides of the table are accurately machined and have T-slots. Workpieces are held on the table with the help of shaper vise, clamps and straps.

### Ram

Ram supports the toolhead on its front. It reciprocates on the accurately machined guideways on the top of the column. It is connected to the reciprocating mechanism placed inside the column. The position of ram reciprocation may be adjusted according to the location of the work on the table.

### Toolhead

The toolhead is fitted on the face of the ram and hold the tool rigidly. It provides vertical and angular feed movement of the tool. The swivel toolhead can be positioned at any required angle and the vertical slide can be moved vertically or at any desired angle to machine vertical or inclined surfaces.

## 3.4 Types of shaping machine

The shaping machines are classified as follows

1. According to the type of driving mechanism
   - a. Crank type
   - b. Hydraulic type
   - c. Geared type
2. According to the design of the table
   - a. Plain shaper
   - b. Heavy duty shaper



XII BME_English version CHAPTER 03.indd 60 1/8/2020 1:43:56 PM